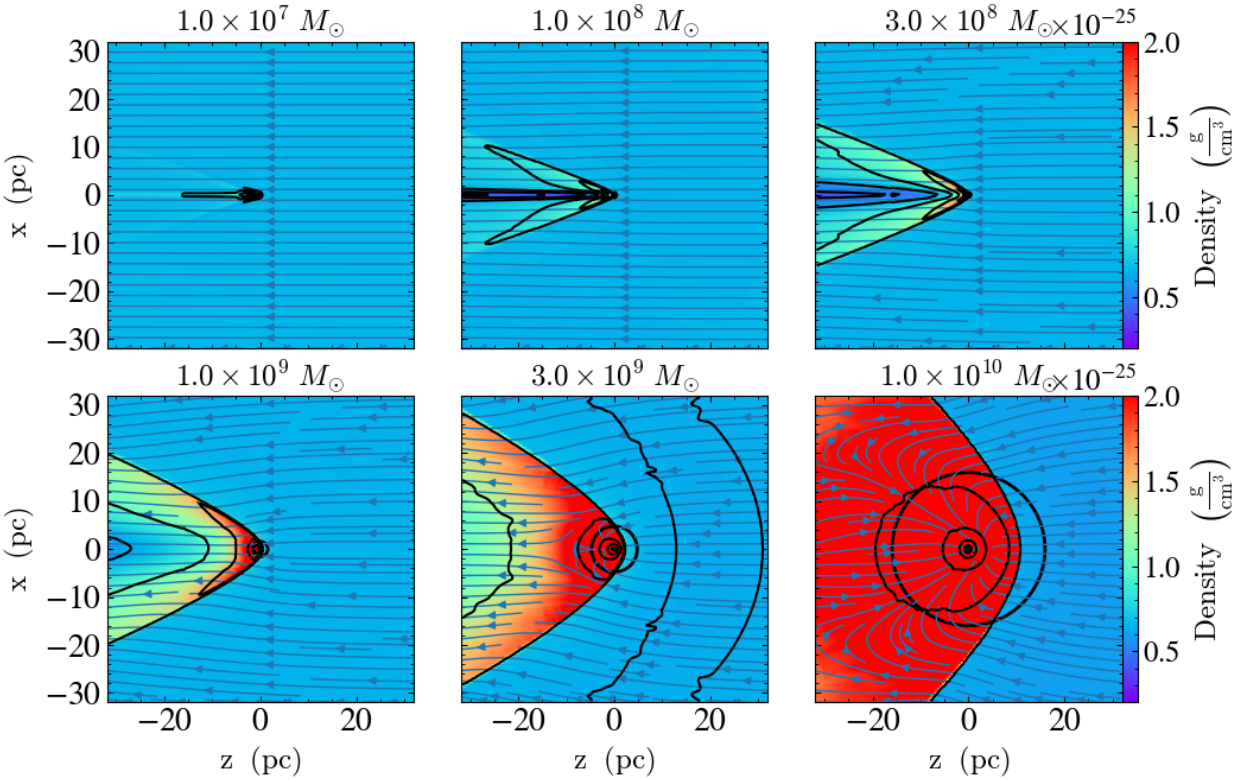

Figure 6. Density slices of the runs with different central black hole masses at $t = 1$ Myr (runs C1 to C6), plotted within a region of size 64 pc. Black circles represent the accretion radius $r_{\rm BHL}$.

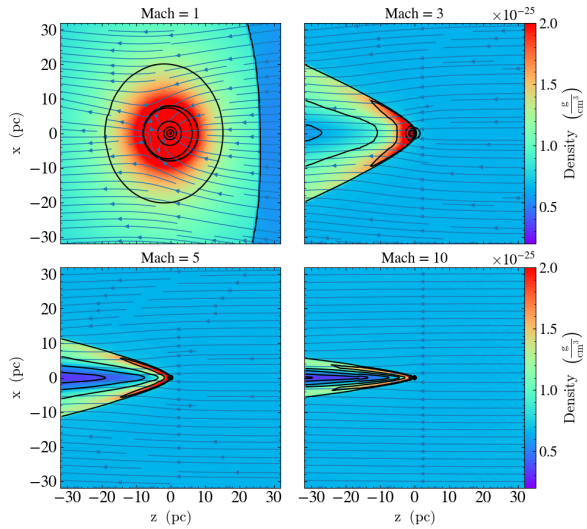

Figure 7. Density slices of the runs with different Mach numbers at $t = 1$ Myr (runs D1 to D4), plotted within a region of size 64 pc. Black circles represent the accretion radius $r_{\rm BHL}$.

what is measured through the control surface inaccurate. The $R_{\rm reset} = 2\Delta x$ case places the bow shock ahead of the resetting region, but produces a larger error in $\dot{M}$, while the $4\Delta x$ case places the shock within the resetting region and produces the most accurate accretion rate. To more easily explore the effects of resolution, we adopt $R_{\rm cs} = 2\Delta x$ and $R_{\rm reset} = 2\Delta x$ for the remaining runs in this section. However, in the jet feedback tests described in § 3.2, we use $R_{\rm cs} = 8\Delta x$ and $R_{\rm reset} = 4\Delta x$.

For the C series runs, we fix $R_{\rm cs}$ and $R_{\rm reset}$ but vary the mass of the black hole, thus changing $r_{\rm BHL}$ from being under-resolved (low mass runs) to well-resolved (high mass runs). The expected behavior of the flow, where the shock forms in front of the accretor, only occurs with well-resolved runs, based on the slice plots shown in Figure 6. Higher resolution leads to a better defined shock, while poorly resolved runs resemble an unperturbed flow more as the wind directly passes the accretor. For the higher mass runs, we measure a standoff distance for the bow shock (the distance between the stagnation point and the closest point on the shock front) to be around $0.2 - 0.3 r_{\rm BHL}$, which is consistent with what Ruffert et al. has observed. Similarly, from the top panel of Figure 9 we observe a more accurate measurement of accretion rate for the well-resolved runs. So long as the control surface and reset radius are similar to or smaller than the BHL radius, the accretion and reset method can capture the flow successfully and reproduce the Bondi-Hoyle-Lyttleton solution to a large extent.

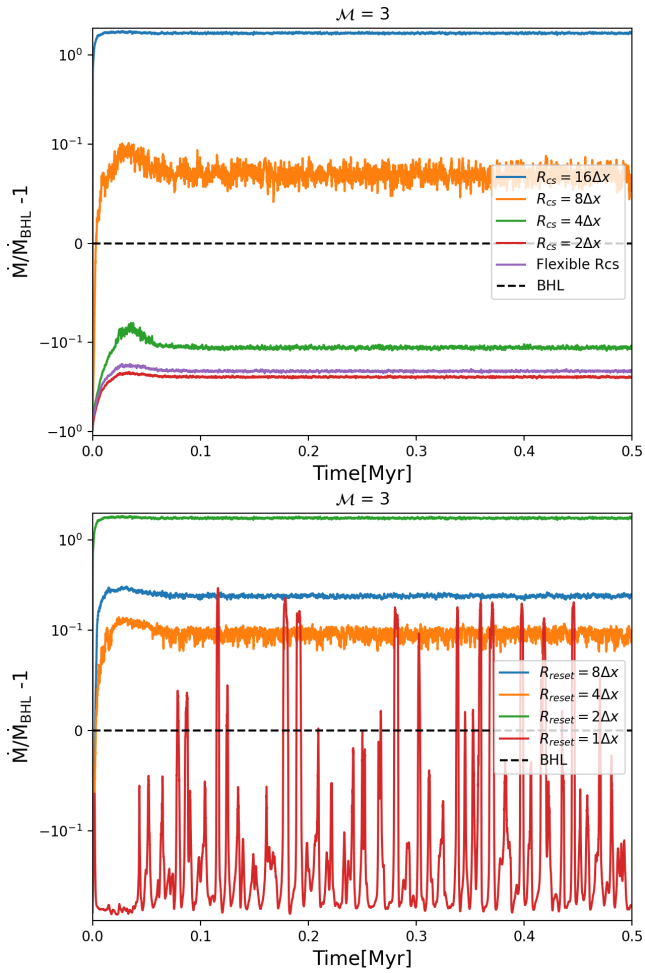

Figure 8. Accretion rate and comparison to the predicted Bondi-Hoyle-Lyttleton rates of different test runs, for A series (top) and B series (bottom).

Run series D, as shown in Figure 7 and the bottom panel of Figure 9, reveals the flow structure for different Mach numbers $\mathcal{M}$. The opening angle of the shock is larger for smaller Mach numbers (with $\theta \sim 20°$ for $\mathcal{M} = 5$ and $\theta \sim 40°$ for $\mathcal{M} = 3$), which is consistent with what has been observed in previous work. The shock stand-off distance also decreases with Mach number, consistent with previous work such as Ruffert & Arnett (1994) and Verigin et al. (2003). This behavior is also consistent with the C series given that higher Mach number makes $r_{\rm BHL}$ much smaller and thus under-resolved in our runs. The convergence behavior with increasing resolution varies for different Mach numbers, however. The best estimate of accretion rate occurs for $\mathcal{M} = 3$, although as seen in the C runs, the error for other runs stays within 10%, possibly due to greater back flow in our simulations. This corresponds to a bow shock jump higher than the other C runs.

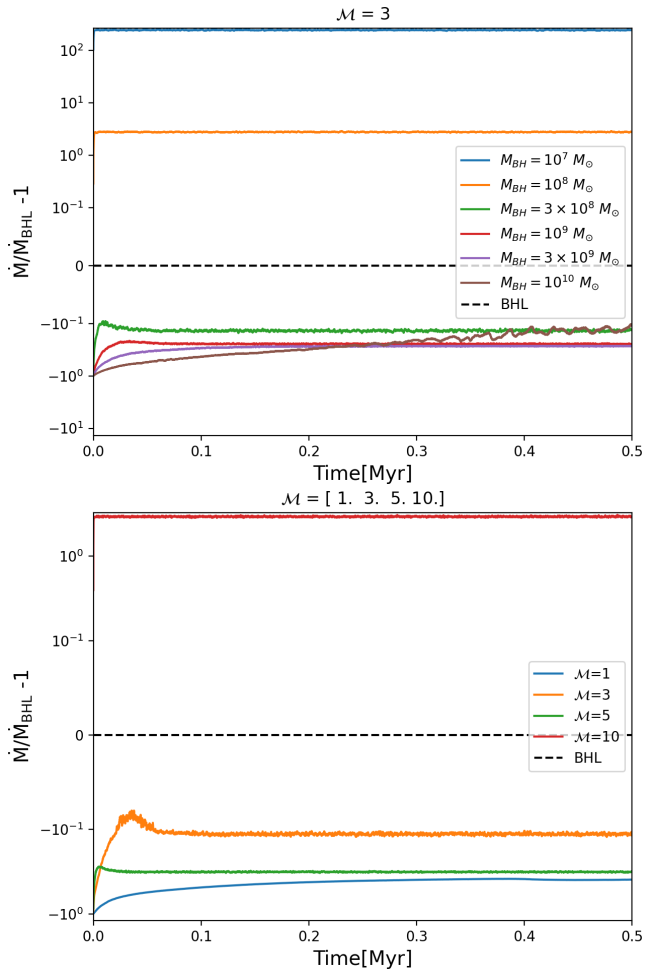

Figure 9. Accretion rate and comparison to the predicted Bondi-Hoyle-Lyttleton rates of different test runs, for C series (top) and D series (bottom).

In general, we show a BHL test with flow solution similar to what has been observed in most previous work, such as Ruffert & Arnett (1994) and Beckmann et al. (2018). The differences arise from the bow shock structure, as our resetting routine is different from what other simulators have used. Accretion rates measured with our method closely match the analytical BHL value, and compared with Beckmann et al. (2018), the dependence of accretion rate on resolution shows a similar behavior. However, we are able to reproduce the expected accretion rate better than their results with similar Mach numbers and resolution configurations.

In many previous works, Bondi-Hoyle-Lyttleton accretion can cause flip-flop instability in the flow inside the shock cone behind the accretor (Taam & Fryxell 1989; Foglizzo & Ruffert 1999). The shock surface oscillates from one side of the accretor to the other, causing the accretion rate to vary in a quasi-periodic behavior. This behavior is observed to be stronger when the accretor is smaller and the flow is more supersonic according to their work. In our test we do not observe a strong flip-flop instability, but we do see some evidence of instability for higher Mach number and more well-resolved runs.

3. JET FEEDBACK METHOD AND TESTS

3.1. *Modeling of the Jet*

Most previous grid-based AGN modeling efforts have implemented feedback, in the form of either jets or bubbles, by adding mass, momentum, and energy source terms to a fixed volume in the vicinity of the accretor (e.g., Yang & Reynolds 2016b; Martizzi et al. 2019). Since we are effectively treating the control surface as an embedded boundary, a more natural approach would seem to be to apply an outward flux on the control surface. A flux-based injection approach is used in dedicated jet simulations and has been used by Brüggen et al. (2007) in the context of AGN feedback. However, in general we must allow for arbitrary numbers of AGN particles with arbitrary spin axes. Since our grid is Cartesian, the spherical control surface does not correspond to a coordinate boundary, and the jet generally emerges at an angle to the grid. The control surface is also usually not well-resolved. Therefore, we have chosen to impose feedback in the traditional way, by adding source terms to zones within a cylindrical volume.

In this and the following sections we use $M_{\rm BH}$ instead of M to refer to the black hole mass.

Following Yang et al. (2012), to impose jet feedback, the mass, momentum, and thermal energy injection rates per unit volume in a given zone are given by

$$\dot{\rho} = \eta \dot{M}_{\rm BH} |\Psi| \tag{22}$$

$$\dot{p} = \sqrt{2\eta\epsilon_{\rm f}(1-\epsilon_{\rm m})}\dot{M}_{\rm BH} c \Psi \tag{23}$$

$$\dot{E}_{\rm th} = \epsilon_{\rm m}\epsilon_{\rm f}\dot{M}_{\rm BH} c^2 |\Psi| \ , \tag{24}$$

where $\epsilon_{\rm f}$ is the feedback efficiency, $\epsilon_{\rm m}$ is the thermal energy factor, η is the mass loading factor, and Ψ is a function of position that describes the jet injection region. For fiducial values, we use $\epsilon_{\rm f} = 0.0625$, $\epsilon_{\rm m} = 0.8$, and $\eta = 2$, making the jet velocity $v_{\rm jet} = 0.112c$, which is mildly relativistic. The jet efficiency value is similar to common values used in other AGN feedback simulations (e.g., Yang & Reynolds 2016a; Li et al. 2017). While these parameters are given constant values in the runs described here, in Paper II we link them to modeling of the black hole accretion disk.

The window function Ψ determines the spatial extent of the jet. Defining cylindrical coordinates (r, z) with the z-axis oriented along the direction of the jet and having its origin at the AGN particle location, we use